Qianyang Chen

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Education

PhD Candidate (Complex Systems) — *University of Sydney, Australia*
School of Computer Science, Faculty of Engineering — *Jul 2023 – Current*

- *Thesis title: Efficiency of collective behaviour in self-organising systems: information-theoretic and thermodynamic perspectives* (Submitted in July 2026)

Master of Engineering (Mechatronics) — *University of Melbourne, Australia*
With Distinction — *Feb 2017 – Dec 2019*

- *WAM: 82.8/100*
- *2019 W.S Robinson Prize – Melbourne School of Engineering*
- *2019 Melbourne Water Women in Water Scholarship*
- *2018 Jack Wynhoven Scholarship*
- *2017 Dean's Honours List – Melbourne School of Engineering*

Bachelor of Actuarial Studies — *Macquarie University, Australia*
With Vice Chancellor's Commendation for Academic Excellence — *Feb 2011 – Dec 2013*

- *GPA: 4.0/4.0*
- *2013 Faculty of Business and Economics Merit Certificate*
- *2012 Ho Kam-chiu Statistics Prize for overall proficiency in 200-level statistics*
- *2012 Department of Economics Prize*
- *2012 Faculty of Business and Economics Merit Certificate*
- *2011 Faculty of Business and Economics Merit Certificate*

Publications

[1] Q. Chen and M. Prokopenko, "Thermodynamic efficiency of self-organization in nonequilibrium steady states," *Physical Review Research*, vol. 8, no. 3, p. 033 265, 2026. DOI: 10.1103/zjjd-gr1b

[2] Q. Chen, N. Ay, and M. Prokopenko, "Generalising thermodynamic efficiency of interactions: Inferential, information-geometric and computational perspectives," *Journal of Physics: Complexity*, vol. 7, no. 2, p. 025 002, 2026. DOI: 10.1088/2632-072X/ae59e1

[3] Q. Chen and M. Prokopenko, "Why collective behaviours self-organize to criticality: A primer on information-theoretic and thermodynamic utility measures," *Royal Society Open Science*, vol. 12, no. 6, p. 241 655, 2025. DOI: 10.1098/rsos.241655

[4] R. Wood, F. King, R. Esmay, Q. Chen, L. Schneider, E. Dotte-Sarout, S. Fallon, K. Fryirs, R. Gillespie, and R. Blong, "The size inherited age effect on radiocarbon dates of alluvial deposits: Redating charcoal fragments in a sand-bed stream, Macdonald River, NSW, Australia," *Radiocarbon*, pp. 1–14, 2023. DOI: 10.1017/RDC.2023.75

[5] R. Blong, K. Fryirs, R. Wood, F. King, L. Schneider, E. Dotte-Sarout, S. Fallon, R. Gillespie, Q. Chen, and R. Esmay, "Inherited age of floating charcoal fragments in a sand-bed stream, Macdonald River, NSW, Australia: Implications for radiocarbon dating of sediments," *The Holocene*, vol. 33, no. 9, pp. 1154–1159, 2023. DOI: 10.1177/09596836231176502

[6] R. Blong, F. Riede, and Q. Chen, "A fuzzy logic methodology for assessing the resilience of past communities to tephra fall: A Laacher See eruption 13,000 year BP case," *Volcanica*, vol. 1, no. 1, pp. 63–84, 2018. DOI: 10.30909/vol.01.01.6384

Workshops & Conferences

- Aug 2026: *SAPoLSN Philosophy Meets the Life Sciences Workshop*. Sydney, Australia.
- Jan 2026: *The Complexity, Criticality and Computation Symposium C^3-2026* (presented talk). Heron Island, Australia.
- Jun 2025: *Santa Fe Institute Complex Systems Summer School*. Santa Fe, NM, United States.
- Feb 2025: *Eleventh International Conference on Guided Self-Organization: Machine Learning in Embodied Agents* (presented talk). Tübingen, Germany.
- Nov 2024: *Centre for Complex Systems Seminar: Disordered System Approaches for Self-Assembling Memristive Neuromorphic Materials*. Sydney, Australia.
- Nov 2024: *Centre for Complex Systems Workshop 2024*. Sydney, Australia.
- Nov 2024: *Workshop on Critical Phenomena and Topological Data Analysis* (presented talk). Newcastle, Australia.
- Oct 2024: *Research Seminar Data Science Foundations* (presented talk via Zoom). Institute for Data Science Foundations, Hamburg University of Technology, Hamburg, Germany.
- Aug 2024: *Emerging Aspirations in Complex Systems Workshop* (presented talk, *Best Presentation Award*). Sydney, Australia.

Work & Project

Mindray Bio-medical Electronics Co., Ltd — *Aug 2021 – Dec 2022*
Data Scientist — *Shenzhen, China*

- Developed predictive models for employee attrition and early-warning systems for medical device failures.

Assistive Lower-limb Exoskeleton (Research & Development) — *Feb 2019 – Nov 2019*
Capstone Project — *Melbourne, Australia*

- Designed control schemes and motion trajectories for assistive exoskeleton.

Peter MacCallum Cancer Centre — *Nov 2018 – Feb 2019*
Research Intern — *Melbourne, Australia*

- Conducted time-series DNA sequencing data analysis and algorithm research.

Aon — *Feb 2014 – Apr 2018*
Actuarial Analyst — *Australia; Singapore*

- Managed reinsurance portfolios, including catastrophe modelling, actuarial pricing and Realistic Disaster Scenario (RDS) analysis.

Teaching & Mentoring

University of Sydney — *Feb 2024 – Current*
Tutor/Associate Lecturer — *Sydney, Australia*

- COMP5048 Visual Analytics, COMP5329 Deep Learning, COMP9123 Data Structures and Algorithms

University of Melbourne — *Jul 2017 – Dec 2017*
Tutor — *Melbourne, Australia*

- ACTL30004 Actuarial Statistics

Macquarie University — *Feb 2013 – Dec 2013*
Peer Assisted Learning Facilitator — *Sydney, Australia*

- AFIN252 Applied Financial Analysis and Management